



M004 Week 5 — English Worksheet

Topic: Your Function Library — Reusable Building Blocks · **Course:** Functions & Games · **Time:** about 45 minutes

This week you collect your best functions into a **library** — a set of reusable building blocks like `build_wall`, `build_tower`, and `clear_area`. Big builds become easy when small functions do the work.

1 · Predict 🧠

Read each set of steps. Before you imagine the build happening, write what you think will happen.

```
function build_wall(length):
    repeat length times:
        place stone block
        move forward 1

build_wall(5)
build_wall(3)
```

The function is called twice with different numbers. How many stone blocks get placed in total? Which call builds the longer wall?


```
function build_tower(height):
    repeat height times:
        place stone block
        move up 1

function build_house():
    build_wall(4)
    build_tower(3)

build_house()
```

build_house calls two helper functions. What gets built, and in what order?


```
function clear_area(size):  
    repeat size times:  
        repeat size times:  
            replace block with air  
            move forward 1  
        move right 1
```

```
clear_area(4)  
build_wall(4)
```

What happens first — clearing or building? Why does the order of these two calls matter?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The library should have **two different** functions: a stone wall and a glass wall. Right now both functions have the **same name**, so the computer gets confused about which one to use.

```
function build_wall(length):  
    repeat length times:  
        place stone block  
        move forward 1  
  
function build_wall(length):  
    repeat length times:  
        place glass block  
        move forward 1  
  
build_wall(4)
```

Hint: every function in your library needs its own clear name.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — `build_house` is supposed to build four walls **and a door**. The door function exists, but the house never gets a door.

```

function build_door():
    replace block with air
    replace block above with air

function build_house():
    build_wall(4)
    build_wall(4)
    build_wall(4)
    build_wall(4)

build_house()

```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — `build_tower(height)` should build a tower **as tall as the number you give it**. But `build_tower(10)` keeps making a tiny tower.

```

function build_tower(height):
    repeat 3 times:
        place stone block
        move up 1

build_tower(10)

```

Hint: the parameter `height` is never used inside the function.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Build something using your own function library — at least one function should call another function. When you finish, come back here.

Send a photo or video of your build, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I made a function called ...

I reused it when ...

One function called another function when ...

I chose the name ... because ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while your code builds. Talk through it like you are teaching someone who has never seen it. Try to use these words: **function**, **parameter**, **call**, **reuse**, **name**.

1. Show your function library before you run anything.
2. Read each function name out loud and say what it builds.
3. Show one place where a function **calls another function**.
4. Say in your own words why reusing functions is better than copying code.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.